Tables for Joins

```
-- Table for Employees
CREATE TABLE employees (
   id INT PRIMARY KEY,
   name VARCHAR(100),
   department_id INT,
   manager_id INT,
   project_id INT
);

-- Table for Departments
CREATE TABLE departments (
   id INT PRIMARY KEY,
   department_name VARCHAR(100)
);

-- Table for Orders
CREATE TABLE orders (
   order_id INT PRIMARY KEY,
   customer_id INT
);

-- Table for Customers
CREATE TABLE customers (
   id INT PRIMARY KEY,
   name VARCHAR(100)
);

-- Table for Students
CREATE TABLE students (
   id INT PRIMARY KEY,
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    name VARCHAR(100),
    course_id INT,
    advisor_id INT
);

-- Table for Courses
CREATE TABLE courses (
    id INT PRIMARY KEY,
    course_name VARCHAR(100)
);

-- Table for Advisors
CREATE TABLE advisors (
    id INT PRIMARY KEY,
    name VARCHAR(100)
);

-- Table for Books
CREATE TABLE books (
    id INT PRIMARY KEY,
    title VARCHAR(100),
    author_id INT
);

-- Table for Authors
CREATE TABLE authors (
    id INT PRIMARY KEY,
    name VARCHAR(100)
);
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```
-- Table for Products
CREATE TABLE products (
    id INT PRIMARY KEY,
    product_name VARCHAR(100),
    category_id INT,
    supplier_id INT
);

-- Table for Categories
CREATE TABLE categories (
    id INT PRIMARY KEY,
    category_name VARCHAR(100)
);

-- Table for Suppliers
CREATE TABLE suppliers (
    id INT PRIMARY KEY,
    supplier_name VARCHAR(100)
);

-- Table for Projects
CREATE TABLE projects (
    id INT PRIMARY KEY,
    project_name VARCHAR(100)
);

-- Table for Players
CREATE TABLE players (
    id INT PRIMARY KEY,
    player_name VARCHAR(100),
    team_id INT
```

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);

-- Table for Teams
CREATE TABLE teams (
   id INT PRIMARY KEY,
   team_name VARCHAR(100)
);

-- Insert data into Departments
INSERT INTO departments (id, department_name)
VALUES (1, 'HR'), (2, 'Engineering'), (3, 'Sales'), (4, 'Marketing');

-- Insert data into Employees
INSERT INTO employees (id, name, department_id, manager_id, project_id)
VALUES
(1, 'Alice', 1, NULL, 1),
(2, 'Bob', 2, 1, 1),
(3, 'Charlie', 3, 2, 2),
(4, 'David', 4, 3, NULL),
(5, 'Eve', NULL, NULL, NULL);

-- Insert data into Orders
INSERT INTO orders (order_id, customer_id)
VALUES (101, 1), (102, 2), (103, 3), (104, NULL);

-- Insert data into Customers
INSERT INTO customers (id, name)
VALUES (1, 'John Doe'), (2, 'Jane Smith'), (3, 'Jim Bean'), (4, 'Anna Baker');

-- Insert data into Students
INSERT INTO students (id, name, course_id, advisor_id)
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VALUES (1, 'Tom', 1, 1), (2, 'Jerry', 2, NULL), (3, 'Sara', NULL, 2);

-- Insert data into Courses
INSERT INTO courses (id, course_name)
VALUES (1, 'Math'), (2, 'Science'), (3, 'History');

-- Insert data into Advisors
INSERT INTO advisors (id, name)
VALUES (1, 'Dr. Stone'), (2, 'Prof. Hill');

-- Insert data into Books
INSERT INTO books (id, title, author_id)
VALUES (1, 'Book A', 1), (2, 'Book B', 2), (3, 'Book C', NULL);

-- Insert data into Authors
INSERT INTO authors (id, name)
VALUES (1, 'Author A'), (2, 'Author B'), (3, 'Author C');

-- Insert data into Products
INSERT INTO products (id, product_name, category_id, supplier_id)
VALUES (1, 'Product A', 1, 1), (2, 'Product B', 2, NULL), (3, 'Product C', NULL, 2);

-- Insert data into Categories
INSERT INTO categories (id, category_name)
VALUES (1, 'Category A'), (2, 'Category B'), (3, 'Category C');

-- Insert data into Suppliers
INSERT INTO suppliers (id, supplier_name)
VALUES (1, 'Supplier A'), (2, 'Supplier B');
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```
-- Insert data into Projects
INSERT INTO projects (id, project_name)
VALUES (1, 'Project Alpha'), (2, 'Project Beta'), (3, 'Project Gamma');

-- Insert data into Teams
INSERT INTO teams (id, team_name)
VALUES (1, 'Team A'), (2, 'Team B'), (3, 'Team C');

-- Insert data into Players
INSERT INTO players (id, player_name, team_id)
VALUES (1, 'Player 1', 1), (2, 'Player 2', 2), (3, 'Player 3', NULL);
```